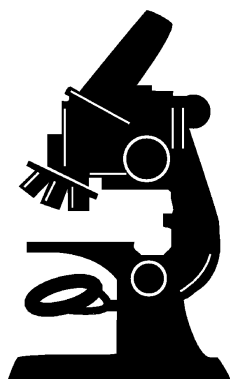




UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational B • 2018



GENERAL DIRECTIONS:

- DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest, and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other**

- B01. All of the following would require an energy source except
- A) endocytosis.
 - B) osmosis.
 - C) exocytosis.
 - D) active transport.
 - E) moving from low concentration to high concentration.
- B02. Ribosomes are composed of two different types of molecules, each belonging to a different organic macromolecular group. To which two groups do these molecules belong?
- A) Protein and Nucleic Acid
 - B) Carbohydrate and Protein
 - C) Lipid and Nucleic Acid
 - D) Carbohydrate and Nucleic Acid
 - E) Lipid and Protein
- B03. In the eukaryotic genome, which structure contains a strand of DNA wrapped around eight histone core proteins?
- A) chromatin
 - B) nucleosome
 - C) 10 nm fiber
 - D) 30 nm fiber
 - E) chromatid
- B04. Millions of acres of rainforest are removed each year in a process called deforestation. Which of the following would not be a potential effect of deforestation?
- A) Reduced gene flow and species fragmentation.
 - B) Increased soil erosion.
 - C) Increased floods.
 - D) Desertification due to decreased atmospheric water vapor.
 - E) Decreased atmospheric carbon dioxide concentration.
- B05. In the Hardy-Weinberg principle, what is the relationship between p and q ?
- A) $p - q = 1$
 - B) $p^2 = q$
 - C) $p + q = 1$
 - D) $p/q = 1$
 - E) $2pq = 1$
- B06. The epithelial tissue lining the _____ contains cells that have microscopic projections, called microvilli, to increase surface area.
- A) lungs
 - B) liver
 - C) bile and pancreatic ducts
 - D) ureters
 - E) small intestines
- B07. An antibiotic that blocks the A-site of the ribosome would immediately
- A) cause dissociation of the ribosome from the mRNA.
 - B) block the binding of incoming tRNAs carrying amino acids.
 - C) prevent transcription.
 - D) cause the mRNA to be misread.
 - E) inhibit ribosomal translocation.
- B08. Which of the following mechanisms can alter allele frequencies within a population?
- A) Gene flow
 - B) Natural selection
 - C) Genetic drift
 - D) Sexual selection
 - E) All of the above alter allele frequencies.
- B09. Polymerase Chain Reaction is a molecular biology technique used to
- A) amplify a segment of DNA, usually from a very small starting quantity.
 - B) insert a plasmid into a bacterial cell.
 - C) determine the sequence of nucleotides on a DNA segment.
 - D) separate nucleic acids or proteins that differ by their size or charge.
 - E) digest DNA at specific palindromic sequences.
- B10. Chickenpox is a disease caused by a/an
- A) bacterium.
 - B) fungus.
 - C) helminth.
 - D) virus.
 - E) protozoan.

B11. What is the maximum theoretical yield of ATP produced per glucose by oxidative phosphorylation alone in aerobic respiration?

- A) 4
- B) 6
- C) 28
- D) 32
- E) 38

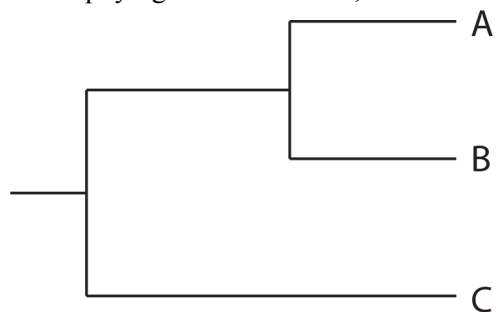
B12. In ecology, an organism that breaks down the remains or waste products of other organisms is called a/an

- A) producer.
- B) scavenger.
- C) consumer.
- D) decomposer.
- E) synthesizer.

B13. A student discovered a complete skull in an open field. After examining the teeth, the student observed that the pre-molars and molars were flat and broad. The student also noticed the canines were absent. Based on this information, the student predicted the skull was mostly likely from a/an

- A) omnivore.
- B) scavenger.
- C) herbivore.
- D) carnivore.
- E) nonmammalian vertebrate.

B14. In the phylogenetic tree below,



- A) A and B are sister taxa.
- B) A and B are unrelated.
- C) C is called a node.
- D) B is the common ancestor to A and C.
- E) C is the common ancestor to A and B.

B15. In theory, how many molecules of carbon dioxide are needed to make two molecules of glucose during photosynthesis?

- A) 1
- B) 3
- C) 6
- D) 9
- E) 12

B16. All of the following are multicellular, eukaryotic parasites except

- A) liver flukes.
- B) tapeworms.
- C) fleas.
- D) malaria.
- E) roundworms.

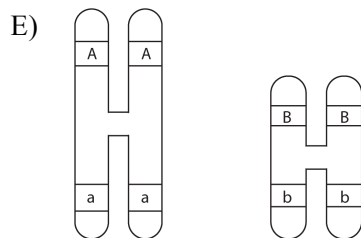
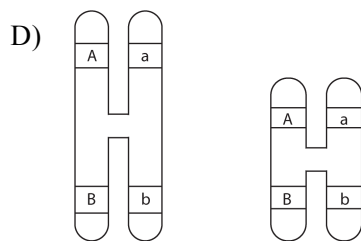
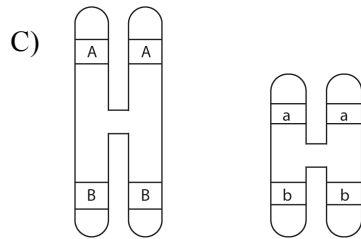
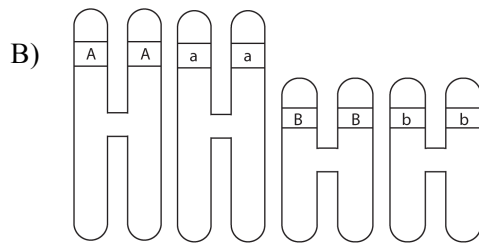
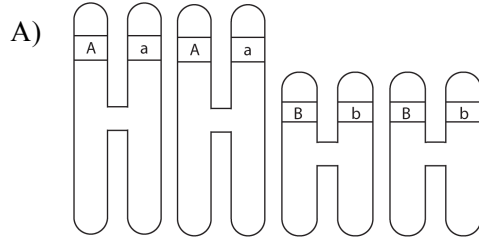
B17. The role of CAP in the regulation of the *lac* operon in some bacteria is to

- A) increase the intracellular concentration of cAMP.
- B) facilitate the binding of the repressor (LacI) to the *lac* operator sequence.
- C) repress the *lac* operon.
- D) assist RNA polymerase in binding to the promoter for the *lacZYA* genes.
- E) prevent translation of *lacZYA* mRNA.

B18. A botanist found a plant that exhibited two dominant phenotypes. Assuming a Mendelian inheritance pattern for both traits, what genotype should the botanist use in a testcross to determine the exact genotype with respect to the two traits of the original plant?

- A) AABB
- B) AaBB or AABb
- C) aabb
- D) AaBb
- E) aaBB or AAbb

B19. An organism with the genotype AaBb has gene A and gene B located on two different chromosomes. Which of the following images is correct for the arrangement of the genes for this organism in G2 of interphase?

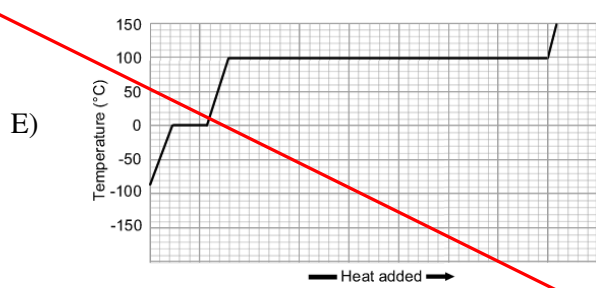
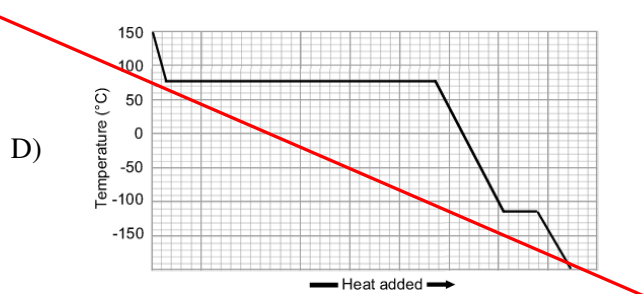
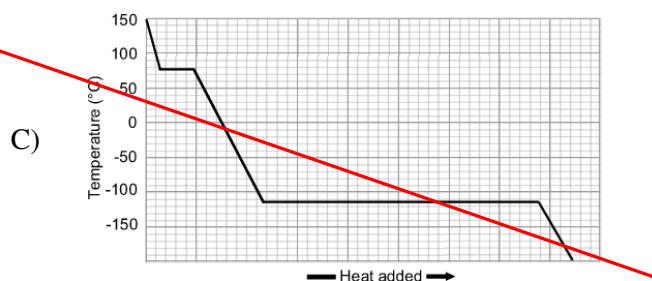
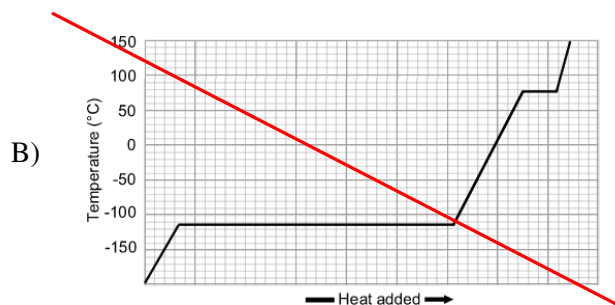
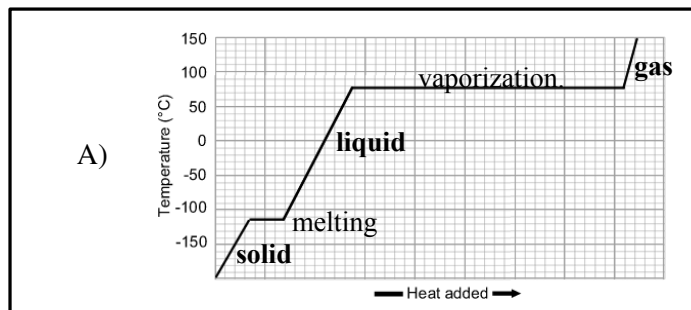


B20. The duck-billed platypus and the echidna are egg-laying monotremes belonging to

- A) Class Mammalia.
- B) Phylum Mammalia.
- C) Kingdom Chordata.
- D) Genus *Homo*.
- E) Domain Animalia.

- C01. Which of the following molecules has sp^3 hybridization on the central atom?
- A) CO_2
 - B) H_2O
 - C) BF_3
 - D) PCl_5
 - E) SF_6
- C02. How many moles of oxygen are there in 5 mol of copper(II) sulfate pentahydrate?
- A) 5 moles
 - B) 9 moles
 - C) 20 moles
 - D) 25 moles
 - E) 45 moles
- C03. If a chemical reaction is described as *spontaneous*, that means the reaction:
- A) happens rapidly
 - B) occurs suddenly, without warning
 - C) results in an overall energy gain
 - D) results in an overall energy loss
 - E) occurs only in the forward direction and cannot be reversed
- C04. Determine the empirical formula of a compound that has a percent composition of 43.7% P and 56.3% O.
- A) P_2O_3
 - B) P_4O_{10}
 - C) P_3O_7
 - D) P_2O_5
 - E) PO_4
- C05. Iron (III) reacts with cyanide ions in water to form $\text{Fe}(\text{CN})_6^{3-}(\text{aq})$. If the initial Fe^{3+} concentration is $1.00 \times 10^{-3} \text{ M}$ and the initial CN^- concentration is 0.150 M and the reaction goes to completion, what will the final concentration of CN^- be?
- A) 0.000 M
 - B) 0.006 M
 - C) 0.144 M
 - D) 0.149 M
 - E) 0.150 M
- C06. Pentaborane, B_5H_9 , was once used as an experimental rocket fuel. It was known as *green dragon* because it explodes on contact with air and burns with a green flame. What is the standard enthalpy change of combustion, ΔH_c° , for pentaborane?
- $$\text{B}_5\text{H}_9 + \text{O}_2 \rightarrow \text{B}_2\text{O}_3 + \text{H}_2\text{O}$$
- A) -1623 kJ/mol
 - B) -3838 kJ/mol
 - C) -4519 kJ/mol
 - D) -7675 kJ/mol
 - E) -9039 kJ/mol
- C07. The yellow line in the helium emission spectrum has a wavelength of 588 nm. What is the energy of one mole of 588 nm photons emitted by helium atoms?
- A) $3.38 \times 10^{-19} \text{ J}$
 - B) $1.48 \times 10^{-28} \text{ J}$
 - C) $1.48 \times 10^{15} \text{ J}$
 - D) $2.04 \times 10^5 \text{ J}$
 - E) $2.04 \times 10^{-4} \text{ J}$
- C08. What is the lattice energy of an ionic compound?
- A) The energy given off when oppositely-charged ions in the gas phase combine to form an ionic bond
 - B) The total bond energy of all the ionic bonds in the crystal
 - C) The difference between the chemical potential of the crystal and the total bond energy
 - D) The energy released when a random arrangement of ions aligns itself into a crystal lattice
 - E) The energy that holds oppositely-charged ions slightly apart from one another in a solid crystal
- C09. How many grams of NaCl would have to be added to 1000 grams of water to raise the boiling point to 102.5°C ?
- A) 39 g
 - B) 78 g
 - C) 120 g
 - D) 140 g
 - E) 280 g

C10. Which diagram below best represents the heating curve for ethanol??



C11. A buffer solution is made in which the concentration ratio of H_2PO_4^- to HPO_4^{2-} is 10 to 1. What is the pH of the solution?

- A) 3.16
- B) 6.21
- C) 7.21
- D) 8.53
- E) 10.0

C12. Which set of quantum numbers below corresponds to the valence electron that is lost when a sodium atom becomes a sodium ion?

- A) $n = 1, \ell = 0, m_\ell = 0, m_s = +\frac{1}{2}$
- B) $n = 2, \ell = 1, m_\ell = 1, m_s = +\frac{1}{2}$
- C) $n = 3, \ell = 0, m_\ell = 0, m_s = -\frac{1}{2}$
- D) $n = 3, \ell = 1, m_\ell = -1, m_s = +\frac{1}{2}$
- E) $n = 3, \ell = 1, m_\ell = 1, m_s = -\frac{1}{2}$

C13. 10 g of nickel(II) iodate is added to 100 mL water. If the equilibrium concentration of iodate ion is $4.55 \times 10^{-2} \text{ M}$, what is the K_{sp} for nickel(II) iodate?

- A) 4.71×10^{-5}
- B) 3.77×10^{-4}
- C) 2.07×10^{-3}
- D) 6.83×10^{-2}
- E) 2.27×10^{-2}

C14. Phosphorus pentachloride decomposes at high temperatures to form phosphorus trichloride and chlorine gas:



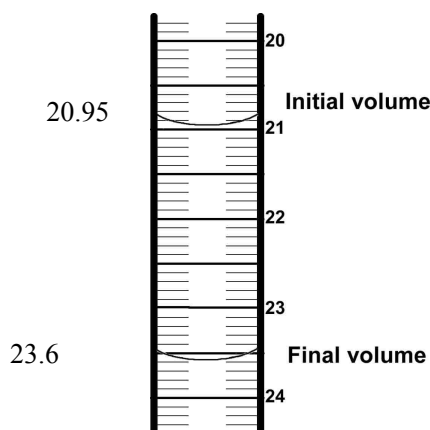
If 1.25 mol of PCl_5 is placed in a 2.0 L container and heated to 425°C and 0.188 mol of PCl_3 is present at equilibrium, calculate the partial pressure equilibrium constant, K_p , for this reaction at 425°C .

- A) 0.0166
- B) 0.579
- C) 0.953
- D) 6.29
- E) 10.1

- C15. An average adult human diet is 2000 Calories per day. One watt is 1 J/sec. Both of these are energy units divided by time. Assuming an average diet of 2000 Calories per day, what is the average wattage of an adult human being?

A) 5811 watts
 B) 1389 watts
 C) 487.3 watts
 D) 96.85 watts
 E) 21.33 watts

- C16. Given the initial and final burette readings in the figure below, what volume was dispensed?



A) 2.5 mL
 B) 2.56 mL
 C) 2.64 mL
 D) 3.36 mL
 E) 5.1 mL

$$23.60 - 20.95 = 2.65$$

- C17. Combustion of ammonia in oxygen produces nitrogen dioxide as one of the products. What is the sum of the coefficients in the balanced equation for the reaction of ammonia with oxygen gas?

A) 4
 B) 8
 C) 12
 D) 16
 E) 21

- C18. The principal ore of lead is lead(II) sulfide, also known by its mineral name, galena. How many tons of lead are in 16.0 tons of galena?

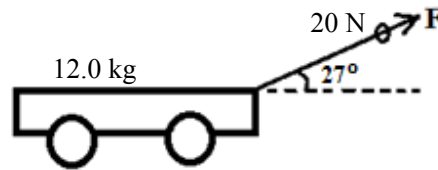
A) 7.43
 B) 10.7
 C) 13.9
 D) 14.4
 E) 14.9

- C19. If a 10.0-liter sample of neon gas at 760 torr is heated from 25.0 °C to 100 °C and compressed to 5.00 liters, what will the final pressure of the gas be?

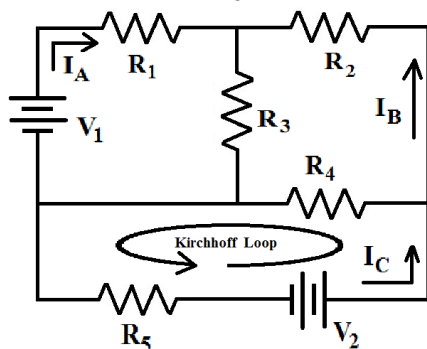
A) 760 torr
 B) 2.50 atm
 C) 8.00 atm
 D) 1.90×10^3 atm
 E) 6.08×10^3 atm

- C20. 100.0 mL of an HCl solution with a pH of 2.00 is added to 100.0 mL of an HCl solution with a pH of 4.0. To the correct number of significant digits, what is the pH of the resulting solution?

A) 2.3
 B) 2.30
 C) 3
 D) 3.0
 E) 3.00

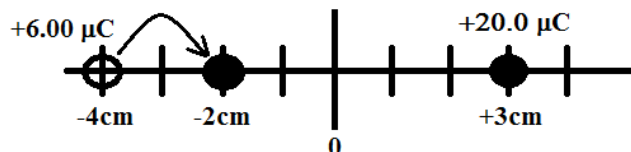
- P01. According to Tyson, scientists know that the Gravitational constant “big G” used in Newton’s equation of gravity has not changed in eons, because if “big-G” had been different in the past then...
- the energy output of the Sun would have been far more variable than the geological record indicates.
 - the orbital mechanics of the Earth would not have permitted a stable circular orbit.
 - the accretion disk around the Sun would have dispersed before planets formed.
 - the Earth’s layers would not have differentiated and the Earth would not have a well-defined core.
 - the spiral arms of the Milky Way galaxy would not have remained stable, and our galaxy would now be an elliptical galaxy.
- P02. According to Tyson, what do astrophysicists use to infer the structure and content of the matter in the early universe?
- the intensity variations of gravitational waves from colliding neutron stars.
 - the temperature variations in the Cosmic Microwave Background.
 - the distribution of small, distant blue spiral galaxies.
 - the distribution of quasars.
 - the calculated intensity fluctuations of dark matter and dark energy.
- P03. According to Tyson, the light from distant quasars and galaxies has been _____ by objects that happen to fall along the line of sight to Earth’s telescopes.
- absorbed
 - spectrally depleted
 - enhanced
 - redshifted
 - gravitationally lensed
- P04. A white dwarf star is primarily composed of what element?
- Carbon
 - Iron
 - Oxygen
 - Helium
 - Hydrogen
- P05. A certain snail you are watching is moving at the exciting speed of 32.0 mm/min. What is the speed of the snail in miles per hour?
- 3.73×10^{-5} mph
 - 4.70×10^{-4} mph
 - 1.19×10^{-3} mph
 - 3.03×10^{-3} mph
 - 1.43×10^{-2} mph
- P06. A golf ball rolls across the roof of a building at 3.00 m/s. It reaches the edge of the roof and flies horizontally off the building at that same speed. The height of the building is 12.50 m. Ignoring air resistance, how far from the base of the building will the golf ball first hit the ground?
- 1.88 m
 - 3.39 m
 - 4.79 m
 - 6.12 m
 - 7.65 m
- P07. A wagon with a mass of 12.0 kg is being pulled as shown by a force of 20.0 N. The force is pulling up at an angle of 27° above the horizontal. Ignoring friction, what is the horizontal acceleration of the wagon?
- 
- The diagram shows a rectangular wagon with two wheels. Above the wagon is the label "12.0 kg". A force vector F is shown as an arrow pointing up and to the right from the front of the wagon. The arrow is labeled "20 N". A dashed horizontal line extends from the point of application of the force, and the angle between this line and the force vector is labeled 27° .
- 0.757 m/s^2
 - 0.849 m/s^2
 - 1.00 m/s^2
 - 1.49 m/s^2
 - 1.67 m/s^2
- P08. A 100.0g ball of clay is thrown directly upwards with an initial velocity of 8.00 m/s. At a height of 2.70m, the clay impacts a 30.0g dragonfly that is hovering with zero velocity. The dragonfly becomes stuck in the clay. What is the speed of the clay + dragonfly immediately after the collision?
- 2.56 m/s
 - 3.33 m/s
 - 5.60 m/s
 - 6.15 m/s
 - 7.27 m/s

- P09. A coin with a mass of 2.50g and a radius of 9.50mm rolls down the entire length of an inclined plane without slipping. The angle of the incline is 45° , and the length of the incline is 84.8cm. Assuming no losses due to friction, what is the linear velocity of the coin at the bottom of the incline? Note: the rotational inertia for a disk is $I = \frac{1}{2}mr^2$.
- A) 2.10 m/s
B) 2.42 m/s
C) 2.80 m/s
D) 3.42 m/s
E) 4.08 m/s
- P10. A simple pendulum has a period of 1.75s. You take the pendulum on an elevator, and while the elevator is accelerating you measure the pendulum's period to be 1.62s. What is the acceleration of the elevator?
- A) 0.728 m/s^2
B) 0.786 m/s^2
C) 1.40 m/s^2
D) 1.64 m/s^2
E) 3.54 m/s^2
- P11. A diatomic gas ($\gamma = 1.4$) starts at a volume of 115cm^3 and a pressure of 3.10atm. The volume is increased adiabatically to 285cm^3 . What is the new pressure?
- A) 0.680 atm
B) 0.870 atm
C) 1.25 atm
D) 1.62 atm
E) 1.80 atm
- P12. For this circuit, determine which loop equation is correct for the Kirchhoff loop indicated. Note that three currents are already labelled.

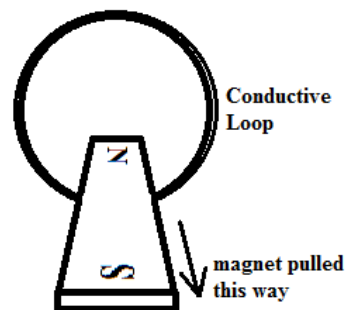


- A) $V_2 - R_4 I_C - R_5 I_C = 0$
B) $V_2 - R_4 (I_C + I_A) - R_5 I_C = 0$
C) $V_2 - R_4 (I_C - I_B) + R_5 I_C = 0$
D) $V_2 - R_4 (I_B - I_C) + R_5 I_C = 0$
E) $V_2 - R_4 (I_C - I_B) - R_5 I_C = 0$

- P13. Determine the energy needed to move a $+6.00\mu\text{C}$ charge from $x = -4.00\text{cm}$ to $x = -2.00\text{cm}$ in this system. There is a $+20.0\mu\text{C}$ charge fixed at $x = +3.00\text{cm}$.



- A) 6.18 J
B) 15.4 J
C) 21.6 J
D) 37.0 J
E) 212 J
- P14. What is the Meissner effect?
- A) The elimination of the magnetic field of a ferromagnetic material by heating.
B) The creation of an aligned magnetic field in a paramagnetic material.
C) The exclusion of a magnetic field from the interior of a superconductor.
D) The deflection of charged particles by a magnetic field.
E) The creation of an electric current by using a changing magnetic field.
- P15. A magnet is arranged with its North pole pointed towards a conductive loop of wire (as shown). If the magnet is quickly pulled away from the loop, in which direction will current be induced in the loop?



- A) clockwise around the loop.
B) counterclockwise around the loop.
C) from bottom to top, up each side of the loop.
D) from top to bottom, down each side of the loop.
E) no current will be induced in the loop.

P16. A concave mirror with a radius of curvature of 60.0cm is hung in a bathroom. If a person stands 1.50m away from the mirror, where does the person's image appear to be located?

- A) 25.0 cm from the mirror
- B) 37.5 cm from the mirror
- C) 42.9 cm from the mirror
- D) 100 cm from the mirror
- E) 158 cm from the mirror

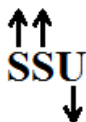
P17. The Balmer lines of Hydrogen are transitions from high energy states down to the $n = 2$ state. The first three Balmer transitions produce photons with wavelengths given in this table:

Transition	Wavelength
$n = 3$ to $n = 2$	656 nm
$n = 4$ to $n = 2$	486 nm
$n = 5$ to $n = 2$	434 nm

Using this data, determine the wavelength of the photon emitted by a hydrogen atom transitioning from $n = 5$ to $n = 3$.

- A) 336 nm
- B) 684 nm
- C) 821 nm
- D) 1280 nm
- E) 2280 nm

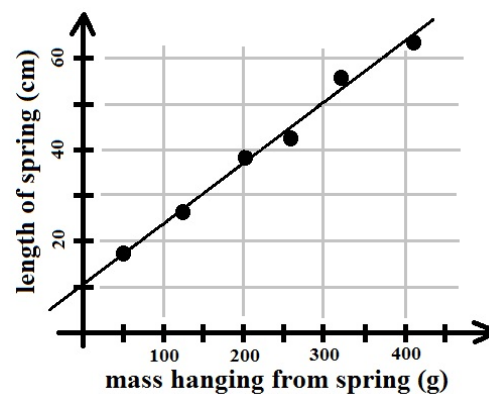
P18. The quark structure of a particle is given by



Which description correctly represents this particle?

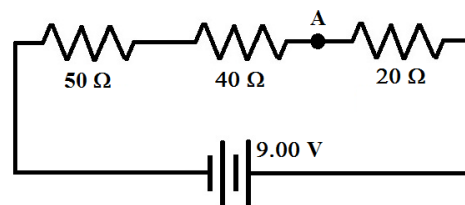
- A) Meson, Charge = 0, Spin = 0, Strangeness = +2
- B) Meson, Charge = +1, Spin = 1, Strangeness = -2
- C) Baryon, Charge = +2, Spin = $\frac{1}{2}$, Strangeness = +2
- D) Baryon, Charge = +1, Spin = $\frac{3}{2}$, Strangeness = -2
- E) Baryon, Charge = 0, Spin = $\frac{1}{2}$, Strangeness = -2

P19. You set up a spring so that it hangs vertically. You then hang various known masses from the spring and measure the length of the spring for each mass. The data you collected is graphed below. What is the approximate spring constant of the spring?



- A) 0.60 N/m
- B) 0.75 N/m
- C) 1.3 N/m
- D) 6.0 N/m
- E) 7.4 N/m

P20. In this circuit, you measure the current at point A to be 129mA. This is definitely not what it should be. What could be wrong with the circuit to explain this measurement? (We assume you are using the ammeter correctly).



- A) The 50 Ω resistor is shorted (zero resistance)
- B) The 50 Ω resistor is open (infinite resistance)
- C) The 40 Ω resistor is shorted (zero resistance)
- D) The 20 Ω resistor is shorted (zero resistance)
- E) The battery is dead (zero volts)

Chemistry																8A 18				
1A 1															2A 2	2				
1 H 1.01															3A 13	4A 14	5A 15	6A 16	7A 17	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18			
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8 9 10			1B 11	2B 12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95			
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80			
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29			
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)			
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)			

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp}	$= 0^{\circ}\text{C}$
T_{bp}	$= 100^{\circ}\text{C}$
c_{ice}	$= 2.09 \text{ J/g}\cdot\text{K}$
c_{water}	$= 4.184 \text{ J/g}\cdot\text{K}$
c_{steam}	$= 2.03 \text{ J/g}\cdot\text{K}$
ΔH_{fus}	$= 334 \text{ J/g}$
ΔH_{vap}	$= 2260 \text{ J/g}$
K_{f}	$= 1.86^{\circ}\text{C}/m$
K_{b}	$= 0.512^{\circ}\text{C}/m$

Constants

R	$= 0.08206 \text{ L}\cdot\text{atm}/\text{mol}\cdot\text{K}$
R	$= 8.314 \text{ J}/\text{mol}\cdot\text{K}$
R	$= 62.36 \text{ L}\cdot\text{torr}/\text{mol}\cdot\text{K}$
e	$= 1.602 \times 10^{-19} \text{ C}$
N_{A}	$= 6.022 \times 10^{23} \text{ mol}^{-1}$
k	$= 1.38 \times 10^{-23} \text{ J/K}$
h	$= 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$
c	$= 3.00 \times 10^8 \text{ m/s}$
R_{H}	$= 2.178 \times 10^{-18} \text{ J}$
m_{e}	$= 9.11 \times 10^{-31} \text{ kg}$

Heat of Formation

Compound	$\Delta H_{\text{f}}^{\circ} \text{ (kJ/mol)}$
$\text{B}_5\text{H}_9(\text{l})$	73.2
$\text{B}_2\text{O}_3(\text{s})$	-1264
$\text{H}_2\text{O}(\text{l})$	-285.8
$\text{H}_2\text{O}(\text{g})$	-241.8

Physical Data for Ethanol

T_{mp}	$= -114.14^{\circ}\text{C}$
T_{bp}	$= 78.24^{\circ}\text{C}$
ΔH_{fus}	$= 4.9 \text{ kJ/mol}$
ΔH_{vap}	$= 38.56 \text{ kJ/mol}$
$c_{\text{liquid ethanol}}$	$= 111.5 \text{ J}/\text{mol}\cdot\text{K}$
$c_{\text{solid ethanol}}$	$= 112.4 \text{ J}/\text{mol}\cdot\text{K}$
$c_{\text{gaseous ethanol}}$	$= 78.28 \text{ J}/\text{mol}\cdot\text{K}$

K_{a} values for Phosphoric Acid

$K_{\text{a}1}$	$= 6.9 \times 10^{-3}$
$K_{\text{a}2}$	$= 6.2 \times 10^{-8}$
$K_{\text{a}3}$	$= 4.8 \times 10^{-13}$

Energy Unit Conversions

1 Calorie	$= 1000 \text{ calories}$
1 calorie	$= 4.184 \text{ J}$

Physics

Useful Constants

quantity	symbol	value
Free-fall acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$
Atomic Mass Unit	u	$1.66 \times 10^{-27} \text{ kg}$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of Sound (at 20°C)	v	343 m/s
Distance Conversion	miles $\rightarrow$ meters	1.00 mile = 1609 meters
Rydberg Constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$